

00:10

Inference Rules

KB1: $(A \vee B) \wedge C \Rightarrow E$

KB2: $D \wedge F \Rightarrow A$

KB3: $\neg E \vee D$

KB4: $B \wedge C$

KB5: F

Prove
A

Propositional Logic
Artificial Intelligence

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Prove Statement using Inference Rules Solved Example – 2

Using forward chaining and rules of inference:

1. C ✓ KB4 and AND-Elimination
2. B ✓ (again) KB4 and AND-Elimination

✓ KB1: $(A \vee B) \wedge C \Rightarrow E$

✓ KB2: $D \wedge F \Rightarrow A$

KB3: $\neg E \vee D$

KB4: $B \wedge C$

KB5: F

✓
Prove A

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Prove Statement using Inference Rules Solved Example – 2

Using forward chaining and rules of inference:

1. C ✓ KB4 and AND-Elimination
2. B ✓ (again) KB4 and AND-Elimination
3. $A \vee B$ ✓ Step 2, using OR-Introduction (introducing A)

✓ KB1: $(A \vee B) \wedge C \Rightarrow E$ -

✓ KB2: $D \wedge F \Rightarrow A$

KB3: $\neg E \vee D$

KB4: $B \wedge C$

KB5: F

✓
Prove A

